

From Synthetic Populations to Pre-Legislative Evidence: A Gated Sim-to-Real Protocol for Regulatory Stress Testing in Argentina

Ignacio Adrián Lerer

Attorney (UBA), Executive MBA (IAE, Universidad Austral).

Buenos Aires, Argentina.

adrian@lerer.com.ar

ORCID 0009-0007-6378-9749

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Abstract

Synthetic populations can expose implementation risks before a proposed rule affects real people, but scale and persona fidelity do not by themselves establish evidence about human behavior. This paper presents an open, gated sim-to-real protocol for pre-legislative regulatory stress testing in Argentina. The protocol freezes the normative text, legislative stage, synthetic cohort, factorial arms, hypotheses, metrics, and acceptance thresholds before independent human observations are collected. A transparent reference implementation tests a fictional national digital-registration duty for micro, small, and medium enterprises across eight combinations of regulatory complexity, enforcement consistency, and procedural safeguards. Four hundred and eighty synthetic personas generate 3,840 deterministic responses. The system reproduces the four encoded directional hypotheses and passes all technical checks, while correctly blocking sim-to-real and legal-policy reliance because no human observations have yet been collected. The contribution is therefore methodological rather than predictive: a reproducible evidentiary architecture that turns uncertainty into workflow state, preserves each legislative version as an immutable wave, and treats synthetic populations as instruments for hypothesis generation and consultation design rather than substitutes for citizens, affected firms, workers, legal authority, or democratic deliberation.

Keywords: synthetic populations; regulatory impact assessment; legislative studies; Argentina; sim-to-real calibration; legal AI; reproducible research

1. Introduction

Legislatures routinely decide under severe informational constraints. A proposed rule may alter incentives, impose unfamiliar procedures, distribute administrative burdens unevenly, or produce avoidance strategies that become visible only after implementation. Regulatory impact assessment seeks to identify such effects early, proportionately, and transparently (Organisation for Economic Co-operation and Development 2020). Yet conventional consultation and field research take time, and legislative texts can change faster than evidence can be gathered.

Synthetic populations appear to offer a shortcut. Language models can produce large samples of persona-conditioned responses, reproduce some established experiments, and support interactive simulations (Argyle et al. 2023; Aher et al. 2023). Recent systems have extended this ambition dramatically. Light Society reports simulations at billion-agent scale (Guan et al. 2025), while MatrAIX introduces 8.3 billion persona records, a one-million-persona coreset, and survey, chatbot, web, and application environments (Li et al. 2026). These systems expand the space of testable scenarios.

Scale, however, does not answer the central evidentiary question: when may a synthetic response support an inference about people? Existing studies document reduced variance, prompt sensitivity, temporal instability, distorted subgroup representations, and the flattening or misportrayal of identity groups (Bisbee et al. 2024; Wang et al. 2025). Persona adherence is not equivalent to human predictive validity. A model may consistently perform a declared persona and still fail to estimate how a relevant population would respond to a legal change. Recent work consequently distinguishes heuristic simulation from statistical calibration with human observations (Hullman et al. 2026).

This paper asks a narrower question: **How can synthetic populations be used before and during legislative deliberation without converting simulation output into premature evidence?** It proposes a gated sim-to-real protocol designed for Argentina’s federal legislative process. The protocol makes four contributions.

First, it separates technical reproducibility, human calibration, and legal-policy reliance into distinct workflow states. Second, it freezes each material version of a proposal as an immutable legislative wave, preventing later changes to the text, model, or thresholds from being mistaken for evidence. Third, it supplies an open reference implementation that runs locally, requires no paid model, and blocks reliance when human observations are absent. Fourth, it locates synthetic simulation within a cumulative research sequence: institutional simulation, cognitively differentiated agents, controlled LLM synthetic societies, and finally prospective human calibration.

The empirical content of this release is intentionally limited. The synthetic run establishes that the harness detects the effects encoded in its transparent response function. It does not validate those effects in Argentina. This limit is a result, not a disclaimer: the system outputs BLOCK for sim-to-real reliance.

2. From Institutional Simulation to Sim-to-Real Evidence

This study does not introduce synthetic legal simulation into the author’s research program. It changes the evidentiary contract under which that simulation may be used. Three prior stages matter.

The first stage modeled reform outcomes directly. *Simulating Institutional Dynamics* implemented a multi-agent engine with seven agent types and nine legal actions, calibrated a Constitutional Lock-in Index against four jurisdictions, and reported Monte Carlo validation against twenty-three Argentine labor reforms (Lerer 2026b). That study treated historical reform outcomes as a validation target. Its contribution was predictive architecture, but its agents remained stylized institutional actors.

The second stage increased cognitive differentiation. *Synthetic Minds in Normative Sandboxes* introduced composite MindProfiles and applied them to a simulated Argentine labor-reform scenario involving one hundred agents, twenty rounds, and five random seeds (Lerer 2026d). It asked how heterogeneous cognitive, affective, and doctrinal profiles alter norm emergence. The framework increased behavioral richness, but it did not establish that simulated profiles predicted an independently observed population.

The third stage treated model families themselves as experimental populations. *Synthetic Populations as Institutional Stress Tests* used TRIBE v2 to expose Claude Haiku 4.5, Qwen 3.7 Max, and Mistral Large 2512 to identical institutional constraints across replicated, stateful and stateless environments (Lerer 2026e). The main result was not a claim about Argentine citizens. It was a model-behavior finding: model families exhibited discontinuous and reproducible institutional response profiles under high constitutional lock-in. That distinction was methodologically important because it stopped treating LLM agents as direct human proxies.

Two adjacent studies expanded the experimental repertoire. A multilevel-selection model of authoritarian persistence combined comparative coding with a stylized agent-based architecture (Lerer 2026a). A preregistered 240-run study of strategic advice compared GPT-4o and Claude Sonnet under controlled prompt variants and identified divergent model phenotypes (Lerer 2026c). Together, these studies established experience with simulation, preregistration, repeated model runs, deterministic judging, and open replication. None, however, supplied a prospective rule for when a simulated regulatory response could become evidence about affected humans.

The present paper addresses that missing transition. Its novelty is not a larger synthetic population, a more psychologically elaborate persona, or a stronger model. It is a gated method for testing whether a frozen synthetic instrument survives contact with independent human observations. This is a methodological advance within an existing program, not a claim of first discovery.

3. Related Work and the Evidentiary Gap

Early work on “silicon samples” tested whether language models conditioned on demographic and attitudinal profiles could approximate subgroup response distributions (Argyle et al. 2023). Aher et al. (2023) proposed Turing Experiments that compare simulated and human results across established behavioral studies. These contributions established a productive research program: synthetic participants can serve as computational objects whose distortions are themselves measurable.

The same literature also identifies serious limits. Bisbee et al. (2024) found that synthetic survey responses could recover some aggregate means while exhibiting too little variation, unstable coefficients, and sensitivity to small prompt changes and time. Wang et al. (2025) showed that replacement-oriented systems can flatten heterogeneity and misrepresent identity groups. These failures matter acutely in legislation, where distributional effects and minority burdens may be more important than an average response.

Population-scale architectures expand coverage but do not remove this gap. MatrAIx reports 91.5 percent persona adherence in a controlled study (Li et al. 2026). That is meaningful evidence that agents can express or suppress assigned attributes. It is not evidence that the same agents predict Argentine workers, firms, unions, lawyers, or citizens facing a proposed rule. Light Society demonstrates engineering scale and structured social transitions (Guan et al. 2025), but simulation fidelity remains task- and population-dependent.

Hullman et al. (2026) sharpen the distinction. Heuristic efforts can improve resemblance, but confirmatory inference requires explicit assumptions and human calibration. The protocol developed here adopts that separation as an operational rule. Synthetic output may identify questions, failure modes, and contrasts worth testing. It cannot cross into behavioral evidence merely because the cohort is large or the personas appear coherent.

4. Institutional Setting and Use Case

Articles 77 through 84 of the Constitution of the Argentine Nation establish the federal process for the formation and enactment of laws (Argentine Republic 1994). Bills may be amended as they move through committees and chambers, and the official congressional description identifies committee reports as a material part of that process (Congress of the Argentine Nation 2026). A useful pre-legislative method must therefore preserve versions rather than treat “the bill” as a stable object.

The reference study uses a fictional rule to avoid implying a position on current Argentine law. The vignette states that a micro, small, or medium enterprise must digitally register and provide advance notice of every modification to working time. The substantive duty remains constant. Three implementation dimensions vary:

1. **Regulatory complexity:** one form and uniform deadline versus multiple forms, variable deadlines, and cross-references.
2. **Enforcement consistency:** public and uniform criteria versus variable criteria across offices and similar cases.

3. **Procedural safeguards:** no consultation, transition, or assistance versus consultation, a ninety-day transition, plain-language guidance, and an assistance channel.

The full factorial design produces eight arms. It tests implementation architecture, not the desirability or constitutionality of the substantive duty.

5. Protocol

5.1 Immutable legislative waves

Each material stage creates a new wave: preliminary draft (T0), formal introduction (T1), committee report (T2), approval by the chamber of origin (T3), final enactment (T4), and implementing regulation (T5). A real wave must record the official URL, retrieval date, source-text SHA-256, and articles in scope. Prior waves are never overwritten.

This design controls a basic source of analytic flexibility. If responses change after committee amendments, the analyst can compare text hashes and experimental artifacts rather than speculate about whether the model or instrument changed silently.

5.2 Frozen study components

Before any human observation, the protocol freezes:

- the normative text and legislative stage;
- the synthetic population and random seed;
- the eight factorial arms;
- four directional hypotheses;
- primary outcomes and action taxonomy;
- calibration metrics and thresholds;
- human sample size, arm minimums, and experience quotas;
- the synthetic response-function version.

The preregistration artifact is canonicalized and hashed. Generated responses receive a separate hash. Altering either artifact breaks reproducibility.

5.3 Synthetic population and response function

The public baseline generates 480 synthetic personas. Each contains four behavioral variables: risk tolerance, decision style, institutional trust, and rule orientation. Values are sampled from fixed categorical supports using seed 20260810. No name, contact detail, party affiliation, voting intention, or direct identifier is generated.

The response function maps only those four behavioral variables and the three experimental factors into five action scores: comply, explore, evade, defect, and abstain. A softmax converts scores to probabilities, and a stable hash of study seed, persona identifier, and arm identifier controls the draw. Comprehension and legitimacy are bounded on the unit interval.

The function intentionally encodes the four hypotheses. High complexity reduces comprehension and compliance and increases evasion and abstention. Inconsistent enforcement reduces legitimacy and compliance and increases evasion and defection. Procedural safeguards increase comprehension, legitimacy, compliance, and exploratory engagement while reducing evasion. This makes the baseline an executable specification, not an empirical behavioral model.

5.4 Independent human phase

The prospective human pilot targets 240 complete responses, thirty per arm. Each participant sees one arm only and cannot view synthetic predictions beforehand. Frozen experience quotas allocate 120 slots to general citizens, sixty to MSME decision makers, thirty to labor or union participants, and thirty to legal or public-policy professionals. The design is a methodological pilot, not a representative national survey.

The instrument records one principal action, a three-item objective comprehension score, legitimacy, feasibility, confidence, experience group, and optional macroregion. It excludes direct identifiers, party affiliation, and voting intention. Human-subject review and consent must be completed before field deployment under the rules applicable to the collecting institution.

5.5 Metrics and gates

For arm a and action k , let p_{ak}^S and p_{ak}^H denote synthetic and human action rates. The action-rate mean absolute error is

$$MAE = \frac{1}{AK} \sum_{a=1}^A \sum_{k=1}^K |p_{ak}^S - p_{ak}^H|.$$

The protocol also computes mean Jensen-Shannon divergence across arm-level action distributions, directional agreement across the four preregistered contrasts, and normalized mean absolute error for comprehension and legitimacy.

The pilot thresholds are MAE at most 0.12, mean Jensen-Shannon divergence at most 0.10, directional agreement at least 0.75, and normalized scale error at most 0.15. At least 240 complete responses and twenty-four observations per arm are required. These are engineering progression gates, not universal statistical standards. They were selected prospectively to decide whether the synthetic model deserves a broader replication.

The workflow states are:

- **Technical PASS:** code, balance, hashes, and metrics operate as specified.
- **Sim-to-real PASS:** a complete human pilot satisfies all frozen thresholds.
- **ESCALATE:** the model fails one or more calibration thresholds, or a legal-policy conclusion is contemplated.
- **BLOCK:** human data are missing or insufficient, artifacts are altered, or prohibited fields appear.

Even a sim-to-real PASS does not authorize legal or policy reliance. Constitutional analysis, fiscal effects, distributive incidence, political support, and population prevalence require separate sources, methods, and reviewers.

5.6 Separation of exploration, confirmation, and reliance

The protocol distinguishes three decisions that are often collapsed. Exploration asks whether a simulated population reveals a plausible failure mode or a discriminating contrast. Confirmation asks whether that contrast is directionally and quantitatively consistent with observations collected from an independently recruited human sample. Reliance asks whether the combined evidence is fit to influence a legal, administrative, fiscal, or political decision.

These decisions require different evidence. A technically reproducible simulation can justify a consultation question. It cannot justify a prevalence estimate. A calibrated pilot can justify continued

investigation. It cannot establish constitutional validity, budgetary incidence, or democratic legitimacy. A legal-policy recommendation requires the relevant authorities, official data, institutional competence, distributional analysis, and accountable human judgment in addition to any simulation evidence.

This separation prevents a common inferential slide:

Stage	Permitted conclusion	Prohibited inference
Synthetic exploration	“This mechanism or failure mode deserves testing.”	“This is how Argentines will behave.”
Human calibration pilot	“The frozen model met or failed prospective calibration thresholds in this sample.”	“The effect has national prevalence.”
Reliance review	“The evidence package is sufficient or insufficient for the specified institutional use.”	“A calibrated model replaces legal authority or democratic deliberation.”

5.7 Legislative-wave operation

The protocol is designed for repeated use as a bill changes. At T0, analysts encode the preliminary draft and identify implementation contrasts. At T1, formal introduction fixes the first official source. Committee reports and amendments create T2 waves only when they alter operative duties, procedures, sanctions, enforcement institutions, or affected populations. Chamber approval, final enactment, and implementing regulation create later waves. Each wave receives its own source hash, preregistration hash, outputs, and disposition.

Comparison across waves is therefore version-aware. If a procedural safeguard added in committee eliminates a modeled evasion mechanism, the result is attributed to a specified textual change. If the same amendment worsens comprehension or shifts burdens to another group, the protocol records the trade-off rather than averaging incompatible versions. This property makes the method useful during legislative deliberation without turning it into an automated vote recommendation.

6. Computational Results

The open implementation generated 480 personas and evaluated each across all eight arms, yielding 3,840 responses. Five unit tests verified deterministic reproduction, balanced human assignments, refusal to calibrate an empty template, rejection of prohibited fields, and detection of artifact tampering. Continuous integration repeats the run under Python 3.10 and 3.12 and fails if generated artifacts drift.

The synthetic contrasts were:

Hypothesis	Outcome and contrast	Synthetic effect
H1	High minus low complexity, evade or abstain	+0.1344
H2	Inconsistent minus consistent enforcement, evade or defect	+0.1234
H3	Safeguards present minus absent under high complexity, comprehension	+0.3000
H4	Safeguards present minus absent under high complexity, legitimacy	+0.3338

All four effects have the encoded direction. The technical state is PASS. The sim-to-real state is BLOCK, complete human (n=0), and the output is not safe for legal or policy reliance.

These results should not be read as estimated effects. They answer a narrower question: does the implementation preserve the design, reproduce the expected mechanism, and refuse to overclaim? It does.

6.1 Reproducibility receipt

The reference run records SHA-256 identifiers for the preregistration, population, and response artifacts. The population and all 3,840 responses regenerate without network access or third-party packages. The committed outputs are checked on two Python versions in public continuous integration. The tests also verify negative behavior: calibration must fail when the human template is empty, forbidden fields must be rejected, and modification of a generated response artifact must be detected.

This receipt supports a claim of computational reproducibility for the public baseline. It does not support construct validity. The response function contains the directional mechanisms that the run recovers. The proper interpretation resembles a software validation test: the implementation behaves according to its specification and maintains the designed evidentiary boundary.

7. Discussion

The central design choice is to make the evidentiary boundary executable. Most warnings about synthetic participants appear as prose attached to an otherwise usable output. A reader can ignore the caveat and retain the number. Here, uncertainty changes the state of the workflow. An empty human template does not yield a weak estimate; it yields BLOCK.

This architecture reframes the role of synthetic populations. Their first function is not substitution. It is disciplined exploration. Before committee debate, a synthetic cohort can identify which implementation features generate the largest model-based contrasts, which questions require human consultation, and where the instrument may lack sensitivity. During debate, immutable waves can test whether amendments remove or create modeled failure modes. After enactment, observed implementation data can challenge both the personas and the response mechanism.

The method also limits demographic overreach. The reference implementation does not map demographic identity to normative behavior. Future studies may need demographic variables to test distributional burdens, but adding them creates a stronger evidentiary and ethical obligation. The flattening and misrepresentation risks identified by Wang et al. (2025) make it unsafe to treat identity prompts as neutral controls.

The protocol complements regulatory impact assessment rather than replacing it. OECD principles emphasize early integration, proportionality, transparency, institutional capacity, and stakeholder engagement (Organisation for Economic Co-operation and Development 2020). Synthetic stress testing can help allocate scarce consultation effort and formulate rival implementation scenarios. It cannot supply the affected parties' lived experience, administrative cost data, legal authority, or democratic legitimacy.

7.1 Why a deterministic baseline is scientifically useful

An LLM-powered public baseline might appear more realistic, but it would make the first validation question harder to answer. Model updates, provider changes, sampling settings, hidden system prompts, and safety policies can alter outputs without a corresponding change in the legal scenario. The deterministic baseline instead exposes every mechanism and produces stable artifacts. It functions as a control instrument against which richer synthetic systems can later be compared.

The baseline also avoids a category mistake. Its four behavioral variables are not claimed to represent Argentine psychology. They are controlled inputs used to test whether the experimental apparatus can

preserve contrasts, balance arms, generate receipts, and stop at the correct evidentiary boundary. Richer agents should be admitted only if they improve out-of-sample human calibration, failure detection, or subgroup coverage relative to this transparent control.

7.2 Strongest objection: calibration may legitimize a bad proxy

The strongest objection is that a small human pilot may confer false legitimacy on a synthetic population. A model could satisfy aggregate thresholds while missing minority burdens, causal mechanisms, temporal changes, or strategically important tails. The protocol mitigates but cannot eliminate that risk.

First, calibration is arm-specific and includes distributional distance, not only average agreement. Second, a PASS authorizes progression, not policy reliance. Third, the human sample and thresholds are frozen before observation. Fourth, later studies must test subgroup and temporal transport rather than assuming them. Fifth, any material change in the model, normative text, population, instrument, or thresholds creates a new version. The method therefore treats calibration as a defeasible property of a defined system under defined conditions, never as a permanent certificate of human equivalence.

7.3 Value during parliamentary treatment

Before formal introduction, the method can compare implementation architectures and identify questions for consultation. During committee treatment, it can document whether amendments address previously modeled failure modes. Between chambers, it can identify new operational contradictions introduced by negotiated text. After enactment, the same frozen outcomes can be compared with administrative and survey data, creating a record of where the prelegislative model succeeded or failed.

The method should not rank political preferences or predict voting intention. Its proper object is implementation architecture: comprehension, feasible compliance, procedural legitimacy, evasion mechanisms, administrative inconsistency, and distributional warning signals. These are legitimate subjects for structured inquiry even when the ultimate legislative choice remains political.

8. Threats to Validity, Limitations, and Falsification

The current study has five decisive limitations.

First, no human observations have been collected. There is no evidence that the synthetic effects approximate responses in Argentina. Second, the public simulator is rule-based rather than LLM-powered. This improves transparency and cost but limits behavioral richness. Third, persona attributes are synthetic and uniformly sampled; they do not estimate the composition of any Argentine population. Fourth, the 240-person prospective sample uses experience quotas and a between-subject design, but it is not a probability sample. Fifth, the thresholds are progression rules, not inferential guarantees.

The protocol is designed to make failure informative. The synthetic model should be revised or rejected if the human phase reverses key directions, exceeds distance thresholds, or reveals subgroup patterns the synthetic cohort suppresses. Any revision must receive a new model version and preregistration hash. Analysts must not tune the current artifact after observing human results and continue calling it the same study.

Internal validity is presently limited by construction: the response function contains the mechanisms being recovered. External validity is absent until human observations are collected. Construct validity remains provisional because comprehension, legitimacy, and action categories compress complex legal behavior. Ecological validity is limited by vignette exposure rather than interaction with an operating rule.

Temporal validity is unknown because institutional trust, political salience, administrative practice, and model behavior can change.

The framework would be falsified in its intended role if prospectively frozen synthetic contrasts repeatedly failed to identify human-observed directions, if a simple non-synthetic baseline performed equally well or better at lower total cost, or if the gates systematically allowed reliance despite missing or contradictory evidence. A more limited negative result would also matter: if richer persona systems failed to improve calibration over the deterministic baseline, their additional realism would be ornamental rather than evidentiary.

9. Research Agenda

The immediate next study is the frozen 240-person pilot. It should report recruitment, exclusions, missingness, arm balance, preregistered metrics, uncertainty intervals, threshold outcomes, and every deviation from the public protocol. The synthetic outputs must remain unavailable to participants and field personnel until data collection closes.

If the pilot fails, the study should publish the failure and diagnose whether it arose from the response mechanism, persona support, instrument, sampling frame, or an invalid underlying hypothesis. If it passes, the next phase should use a larger and more diverse sample, hold out at least one legislative scenario, compare the deterministic baseline with one or more LLM-based synthetic populations, and measure incremental value against simpler methods such as expert elicitation or ordinary cognitive pretesting.

Only after those stages should the program consider an audited subset of a population-scale persona resource. The relevant criterion is not the number of available personas. It is whether a provenance-bearing subset improves detection of material failures or coverage of affected groups without increasing distortion, privacy risk, or false precision.

10. Conclusion

Synthetic populations can make legislative analysis faster only if they also make its limits harder to ignore. The protocol presented here links every modeled result to a frozen text, wave, cohort, mechanism, and gate. It uses synthetic agents to propose and stress-test hypotheses, then requires independent human observations before behavioral claims can proceed. It keeps legal and policy reliance behind a separate boundary.

The open reference implementation has completed its first task: it reproduces the encoded mechanisms and blocks sim-to-real inference. Read alongside the author's earlier institutional simulations, cognitive-agent sandbox, and TRIBE v2 model-population experiments, the result marks a change in research direction. The problem is no longer how to make a synthetic society larger or more elaborate. It is how to determine, prospectively and audibly, whether any such society earns a limited evidentiary role.

The next discriminating test is therefore not a larger synthetic population. It is the preregistered human pilot. Until that evidence exists, the correct output is not a prediction with a caveat. It is BLOCK.

Data, Code, and Reproducibility

The code, preregistration artifact, synthetic personas, synthetic responses, human-response template, tests, and generated receipts are available at <https://github.com/adrianlerer/prelegislative-sim-to-real-argentina>. The reference run requires Python 3.10 or later and no third-party package or network access.

Ethics Statement

No human participants or personal data were used in the reported computational run. The supplied human instrument is prospective and must not be deployed until the collecting researcher or institution completes the applicable ethical, consent, and data-governance review. Synthetic responses must never be inserted into the human template and represented as observed data.

AI Assistance Disclosure

AI tools assisted with code generation, drafting, source discovery, and consistency review under the author's direction. The author selected the research question, architecture, claims, sources, and evidentiary boundaries; verified the cited metadata and executable outputs; and accepts responsibility for the manuscript and release.

Competing Interests

The author develops legal-technology and AI-governance systems. The public replication package excludes private product logic, client data, proprietary prompts, model-routing policies, and commercial evaluation thresholds. No external funding was received for this study.

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